CMSC335

Web Application Development with JavaScript

CSSI

Department of Computer Science

University of MD, College Park

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Announcements

- Please read the syllabus again ☺
- Make sure you have more than one device for authentication
- For transfer students
  - TCC (Transfer Connect Club) is a social club for transfer students focused on integrating transfer students into the UMD community. All students, even non-transfer ones, are welcome.
  - Discord: https://discord.gg/PMquS9M7aQ
  - Contact: Hannah, email: hrc00017@terpmail.umd.edu

Miscellaneous

- **VS Code Settings**
  - Select the wheel (bottom left) and then "Settings“
  - Auto Save with **afterDelay** option automatically saves every second
    - » Recommended
  - The activity bar position can be changed to bottom, top, hidden, default
    - » Right-click on the bar and select “Activity Bar Position”
- **Hard Refresh/Hard Reload** - clears browser cache for a specific page, forcing a loading of the most recent version of the page. Note: "Refresh" arrow on your browser address bar is not a hard refresh
  - Windows - hold down Ctrl, Shift, and press R
  - Mac - hold down Cmd, Shift, and press R
- **Soft Refresh** - reloads the page from the cache
- **Commenting out code in VS Code**
  - Ctrl and forward slash turns a line into a comment (or vice versa). Works for HTML, CSS, JavaScript
- Above information also available at
- https://www.cs.umd.edu/~nelson/classes/resources/web/ (under IDE/Editors)

CSS (Cascading Style Sheets)

- Official W3C standard **for controlling presentation**
- Style Sheets
  - Text file with rules
  - They include no HTML
  - Use a **.css** extension
  - Allow you to apply typographic styles (font size, line spacing, etc.)
  - Allow you to have page layout control
  - Allow smaller HTML files by avoiding redundancy in style specification
  - Allow easy update of a collection of pages by updating only a single file
- Why CSS?
  - http://www.csszengarden.com/

CSS

- Rule
  - Basic element of a style sheet
  - **Example:** h1 {color: blue; text-align: center;}
  - **Describes the formatting associated with a page element**
- Rule format

  selector { declaration block }
- **Selector**: identifies what should be styled in a web document
  - e.g., h1, p
- **Declaration**
  - Describes styling information (what and how that portion of the web document should be modified)
  - **Example:** color: blue
  - **Declaration format:** property (e.g., color), colon (:) and value (e.g., blue)
- **Declaration block:** declarations enclosed in { } and separated by semicolons (last one does not need a semicolon, but it is OK to have it)
- **Properties list:** http://www.w3schools.com/cssref/

Types of Style Sheets

- **Inline**
  - Style information applied to a specific tag
    (e.g., New Semester)
  - Avoid if possible
- **Internal**
  - Using the <style> tag in the header of the HTML document
  - Convenient to provide own style to a specific page
    - » It can override an include style sheet
  - **Example:** InternalStyle.html
- **External**
  - External style sheet which web pages link to (see <link> tag)
  - Preferred approach
- **Example:** ExternalFile.html, ExternalFile.css
  - Files do not need to have the same base name
- We can use /* multi-line */ to comment CSS

CSS Validator

- http://jigsaw.w3.org/css-validator/
- Notice you have three choices
  - By URI
  - By file upload
  - By direct input
- Look at the “More Options” section
- We can also validate using the VS Extension you installed for HTML
  - Let’s see a demo

Regarding Size

- Two types of length units:
  - **Relative** (e.g., em, rem, %)
    - » **Example:** 3em
    - » **Example:** 4rem
  - **Absolute** (e.g., px, cm, pc)
- For the course: we will mainly use pixels, em, rem, %
- Relative
  - **em**: Relative to the size of the current font
    - » 3em: 3 times the size of the current font
  - **rem ("root em")**: inherits its value from the base font size of the root element
    - » In HTML, the root element is the <html> element
    - » **rem** is constant throughout a document
  - **Examples:** SizeRemVsEm.html
  - The **em** and **rem** units are practical in creating a scalable layout
  - http://www.sitepoint.com/power-em-units-css/
- Whitespace cannot appear between the number and the unit
  - correct: 3em, incorrect: 3 em
  - If the value is zero, the unit can be omitted

Regarding Colors

- You can specify colors using predefined colors (e.g., **yellow, white, red, purple, blue**)
- You can specify a color by indicating the red, green, and blue components. For example, all the following are equivalent:
  - red
  - rgb(255,0,0)
  - #ff0000 or #f00 (shorter, cannot use it for all colors)
- RGBA adds an alpha channel that allows us to specify the opacity for a color
  - RGBA color specification: **rgba(RED, GREEN, BLUE, ALPHA)**
  - Alpha value of 0 (transparent), alpha value of 1.0 (opaque)
    - » **Example:** rgba(125, 125, 125, 0.5);
- In VS Code, you can define a particular color (including opacity). Hover over a color of a CSS declaration in VS Code to see options
- References:
  - http://www.w3schools.com/css/css3_colors.asp

Kinds of Selectors

- **Type Selectors**:
  - Based on the name of an HTML tag (e.g., p, table, etc.)
  - **p** { color: red; }
- **Class Selectors**:
  - Allow us to define a style ("type of style")
    - » Allow us to apply the same rules to a set of elements
  - Created with a period **.** (also known as a full stop)
  - Applied to an element using the **class** keyword
  - You can apply several class selectors to an element
  - **Example:** ClassIdSelectors.html
- **ID Selectors:**
  - Similar to class selectors but appear only once in the document
  - Used when you need to apply a style only once in your document, or you want to capitalize on an id that already exists
  - Created using **#**
- **Example:** ClassIdSelectors.html

Kinds of Selectors

- **Pseudo-class** - "phantom class" that allows assigning style to elements based on the **state of an element** (e.g., a link has been clicked) or markup patterns (e.g., odd rows of a table)
  - Attached to selectors (e.g., p:hover)
  - Four popular pseudo-classes uses are
    - » **a:link**: the initial color of a link
    - » **a:visited**: color for a visited link
    - » **a:hover**: color when the mouse hovers over the link
    - » **a:active**: color during the clicking of the link
- **Example:** Pseudoclass.html
  - After selecting a link, you can restore the initial link color by choosing (in Chrome) History, History (yes, twice), Delete browsing data, and selecting Browsing history. Uncheck any options (e.g., Cookies and other site data) you may want to keep

About Properties

- Inheritance of Properties:
  - A child element inherits the same properties of its parent element (unless otherwise indicated by the CSS specification)
  - **Examples of inherited properties**
    - » color, font-family, font-size, letter-spacing, …
  - **Examples of non-inherited properties**
    - » border, margin, padding, …
- Properties can overwrite others
  - **Example:** Overriding.html
- " Cascading" in Cascading Style Sheets
  - Determines the final style for an element when multiple can apply. The “cascade” refers to combining different style sources

font-family

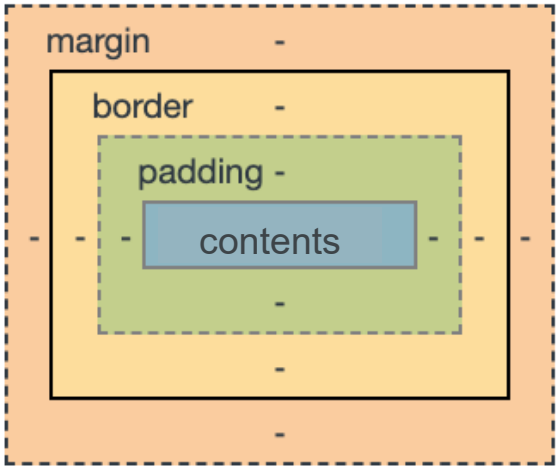
- In a document we usually associate with the <body> tag a font-family
- The font-family property defines the fonts to use (in order)
- Two generic font families used as default values: serif and sans-serif
  - **serif**
    - » Line or stroke added to letter or symbol
    - » **Examples:** Times New Roman, Georgia
  - **sans-serif**
    - » No serif (no line or stroke added to letter or symbol)
    - » **Examples:** Verdana, Arial
- Additional generic font families
  - **monospace**
    - » **Example:** Courier New, Consolas
  - **cursive** - *Lucida Handwriting*
  - **fantasy** - Comic Sans MS, …
- **Example:** FontFamily.html

Lorem Ipsum

- Lorem ipsum - placeholder text that serves to fill a space without actually saying anything
- **The main reason for its use is so people don't focus on the text**
- It is rooted in Latin
- Comes from sections 1.10.32 and 1.10.33 of "de Finibus Bonorum et Malorum" (The Extremes of Good and Evil) by Cicero
- Reference: https://www.lipsum.com/
  - You can generate Lorem ipsum lists, paragraphs, etc., using the previous site
- Use <html lang="zxx"> (HTML validation my fail otherwise)

Box Model

- CSS box model:
  - Box around an HTML element with the following components: margin, border, padding, content
- Allow us to define space between different elements
- Four sections can be identified:
  - **contents**: what lies in the middle of the box (text, image, etc.)
  - **padding**: surrounds the content
  - **border**: surrounds the padding and represents the box border
  - **margin**: surrounds the border



Box Model (Padding/Margin)

- You can adjust individual padding/margin properties by using:
  - padding-bottom, padding-left, padding-top, padding-right
  - margin-bottom, margin-left, margin-top, margin-right
- The margins, borders, paddings, and background properties of block elements (e.g., body, p, etc.) are not passed to the children
  – i.e., not inherited
- Box width
  - left + right padding, left + right border, left + right margin, content width
  - **width** property - sets the **content** width
- Box height is determined in the same way
  - **height** property - sets the **content** height
- **Example:** BoxModel.html

Shorthand Property

- Allows you to specify several properties by using only one
  - If you don’t specify one of the properties, a default value will be used
- Commonly used shorthand properties
  - background
  - font
  - list-style
  - margin
  - border
  - padding
- **Example:** ShorthandProperties.html

Background properties

- Background properties can be applied to <p>, , <body>, and other elements
- **background-color** - color
- **background-image** - location (URL) of image
- **background-repeat** - how the image repeats
  - Possible values for repetition
    - » **no-repeat** - one instance of the image
    - » **repeat** - tile
    - » **repeat–y** - repeats on the y-axis
    - » **repeat–x** - repeats on the x-axis
- **background-attachment**
  - Indicates attachment of the image to the containing element
  - Possible values are:
    - » **scroll**: default value
    - » **fixed**: the image will stay stationary as the scrolling takes place

Background (cont.)

- **background-position**
  - Possible values (some combinations are valid)
    - » top, bottom, center, left, right
- **background-size**
  - Possible values
    - » auto auto - retains the original size
    - » 000px 000px
    - » x% y%
    - » contain | cover | …
- There is a shorthand property for backgrounds
  - background: lightblue url("campusBldg.jpg") no-repeat fixed center;
- **Examples:** Background.html

Formatting Lists and Tables

- **Example:** ListsFormatting.html
- **Examples:** TablesFormatting1.html, TablesFormatting2.html